R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

Second Year (Semester – 3)

Course Content & Grade

Course Code:

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Advances FEM and Programming**	**MTSE-3001(A)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Analyze the various structural members using plate, shell and hybrid elements.

CO2: Analyze dynamic problems and vibration of bars, beams and plate elements using FEM

CO3: Analyze the buckling of struts and plate elements using FEM.

CO4: Analyze shear walls, core walls, bridges and cooling towers using FEM.

CO5: Understand the computational aspects and interpretation of results of FEM and also compare with other methods.

Course Contents

Unit 1

Iso-parametric formulation for plate and shell elements; various types of elements; Hybrid elements.

Unit 2

FEM in dynamic problems, consistent mass matrix; Vibration of bars, beams and plate elements.

Unit 3

FEM in buckling problems, geometric matrix, buckling of struts and plate elements.

Unit 4

Structural modeling by FEM for structures such as shear walls, core walls, bridges and cooling towers.

Unit 5

Computational aspects; interpretation of results; comparison with other methods.

Reference Books

1. Weaver, Johnson, Finite element and structural analysis
2. H. C Martin, Matrix structural analysis
3. C.F. Abel, C.S. Desai, Finite element methods, CBS Publishers New Delhi
4. Buchanan, Finite element Analysis (Schaum Outline S), Tata McGraw Hill
5. Krishnamurthy, Finite element analysis, Tata McGraw Hill
6. Zinkiewicz, O.C. and Taylor, R.L., The Finite Element Method, McGraw-Hill
7. Reddy, J. N., An Introduction to Finite Element Method, McGraw-Hill, Singapore

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Design of Steel Structure**	**MTSE-3001(B)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the concept of Limit state method in design of steel structures.

CO2: Analyze and design the various types of columns.

CO3: Analyze and design the laterally restrained beams

CO4: Analyze and design beam columns.

CO5: Analyze and design beams subjected to torsion and bending

Course Contents

Unit 1: Introduction to Limit States

Introduction, Standardization, allowable stress design, limit state design, partial safety factors, concept of section, classification; Plastic, compact semi-compact & slender.

Unit 2: Columns

Basic concepts, strength curve for an ideal strut, strength of column members in practice effect of eccentricity of applied loading. Effect of residual stresses, concept of effective lengths, no sway columns, torsional and torsion flexural buckling of columns, Robertson's de sign curve, modification to Robertson approach, design of columns using Robertson approach.

Unit 3: Laterally Restrained Beams

Flexural & shear behavior, web buckling & web crippling, effect of local buckling in laterally restrained plastic and compact beams, combined bending & shear, unsymmetric al bending.

Unrestrained Beams: Similarity of column buckling of beams, lateral torsional buckling of symmetric section, factors affecting lateral stability, buckling of real beams, design of cantilever beams, continuous beams.

Unit 4: Beams Columns

Short & long beam columns, effects of slenderness ratio and axial force on modes of failure, beam column under biaxial bending, strength of beam columns, local section failure & overall member failure.

Unit 5: Beams subjected to Torsion and Bending

Introduction, pure torsion and warping, combined bending torsion, capacity check, buckling check, design methods for lateral torsional buckling.

Reference Books

1. Morsis L.J. Plum, D.R., Structural Steel Work Design, Nichols Pub Co.

2. B. S. Krishnamachar and D. Ajitha Sinha, Design of steel structures Tata McGraw Hill Publishing Co. Delhi.

3. Yu, W.W., Cold Formed Steel Structures Design, John Wiley & Sons; 4th edition, 2010.

4. Subramanian N, Design of Steel Structures, Oxford University Press, New Delhi, 2013.

5. Narayanan R.et.al., Teaching Resource on Structural Steel Design, INSDAG, Ministry of Steel Publications, 2002

6. Duggal S. K., Limit State Design of Steel Structures, Tata McGraw Hill Publishing Company, Third edition, 2019.

7. Bhavikatti S. S, Design of Steel Structures by Limit State Method as per IS:800-2007, IK International Publishing House Pvt. Ltd., 2009

8. Shah V. L. and Veena Gore, Limit State Design of Steel Structures, IS 800-2007, Structures Publications, 2009.

9. IS 800: latest version, General Construction in Steel - Code of Practice, Bureau of Indian Standards, New Delhi.

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Design of Tall Structure**	**MTSE-3001(C)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the lading and design principles for tall structures.

CO2: Understand the behavior of various structural systems

CO3: Analyze and design tall structures using various methods.

CO4: Understand the various structural elements

CO5: Analyze and design various other tall structures such as chimney, TV towers etc.

Course Contents

Unit 1 Loading and Design Principles

Loading- sequential loading, Gravity loading, Wind loading, Earthquake loading, Gust Factor and Karman Vortices. Approximate and Regorlons Methods of analysis for wind and Earthquake Forces, combination of loading, Static and Dynamic approach, Analytical and wind tunnel experimental methods, Design philosophy, Working stress method, Limit state method and Plastic design.

Unit 2 Behavior of Various Structural Systems

Factors affecting growth, Height and structural form, High rise behaviour, Rigid frames, braced frames, Infilled frames, Shear walls, Coupled shear walls, Wall-frames, Tubulars, cores, Outrigger-braced and hybrid mega systems.

Unit 3 Analysis and Design

Modeling for approximate analysis, Accurate analysis and reduction techniques, Analysis of buildings as total structural system considering overall integrity and major subsystem interaction, Analysis for member forces, drift and twist, Computerized three dimensional analysis, Assumptions in 3D analysis, Simplified 2D analysis.

Unit 4 Structural Elements

Sectional shapes, Properties and resisting capacity, Design, Deflection, Cracking, Prestressing, Shear flow, Design for differential movement, Creep and shrinkage Effects, temperature and fire resistance, Ductility and reinforcement details at joint.

Unit 5 Other Tall Structures

Criteria for design of Chimneys, T.V. Towers and other Tall Structures, Case studies.

Reference Books

1. Bryan Stafford Smith and Alexcoull, Tall Building Structures - Analysis and Design, John Wiley and Sons, Inc., 2005.
2. Taranath B.S., Structural Analysis and Design of Tall Buildings, McGraw Hill, 2011.
3. Wolfgang Schueller, High Rise Building Structures, John Wiley and Sons, New York 1977.
4. Beedle.L.S., Advances in Tall Buildings, CBS Publishers and Distributors, Delhi, 1986.
5. Lin T. Y, Stotes Burry. D, Structural Concepts and systems for Architects and Engineers, John Wiley, 1988.
6. IS 875 (Part 1): latest version, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures Part 1 Dead Loads - Unit Weights of Building Materials and Stored Materials, Bureau of Indian Standards, New Delhi.
7. IS 875 (Part 2): latest version, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures Part 2 Imposed Loads, Bureau of Indian Standards, New Delhi.
8. IS 875 (Part 3): latest version, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures Part 3 Wind Loads, Bureau of Indian Standards, New Delhi.
9. IS 875 (Part 4): latest version, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures Part 4 Snow Loads, Bureau of Indian Standards, New Delhi.
10. IS 875 (Part 5): latest version, Indian Standard Code of Practice for Design Loads (Other than Earthquake) for Buildings and Structures Part 5 Special Loads and Combinations, Bureau of Indian Standards, New Delhi.
11. IS 1893 (Part 1): latest version, Indian Standard Criteria for Earthquake Resistant Design of Structures, Part 1 General Provisions and Buildings, Bureau of Indian Standards, New Delhi.
12. IS 4998-1 (1992): Criteria for design of reinforced concrete chimneys, Part 1: Assessment of loads, Bureau of Indian Standards, New Delhi.
13. IS 6533-2 (1989): Code of practice for design and construction of steel chimneys, Part 2: Structural aspects, Bureau of Indian Standards, New Delhi.
14. IS 11504 (1985): Criteria for structural design of reinforced concrete natural draught cooling towers, Bureau of Indian Standards, New Delhi.
15. IS 13920 (2016): Ductility Design and Detailing of RC Structures subjected to Seismic Forces, Bureau of Indian Standards, New Delhi.

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Stability Theory in Structural Engineering**	**MTSE-3001(D)**	**3L-1T-0P**	**4**

Course Outcomes

After studying this subject, the students will be able to:

CO1: Understand the concept of stability theory.

CO2: Analyze torsional buckling problems.

CO3: Analyze lateral instability of beams.

CO4: Analyze the buckling of plates.

CO5: Analyze the stability problems using energy and matrix methods

UNIT 1

Concepts of Stability, Euler Buckling Load, Critical Load of Laced, Battened and Tapped columns, Inelastic Buckling of column.

UNIT 2

Torsional Buckling, Torsional Flexural Buckling.

UNIT 3

Lateral Instability of Beams, Beam Columns.

UNIT 4

Local Buckling and post buckling behaviour of plates.

UNIT 5

Application of Energy method and matrix method in stability problems.

Reference Books

1. Timoshenko, S. P. & Gere, James M. Theory of Elastic Stability (2nd. ed.). McGraw –Hill International Editions, Mechanical Engineering Series.
2. Iyenger, N.G.R. Structural Stability of Columns & Plates. Ellis Horwood Ltd, Publisher.
3. Principles of Structural Stability Theory by Alexander Chajes, Waveland Pr Inc

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3**)

Course Content & Grade

Course Code:

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	SWAYAM/NPTEL-MOOC-1	MTSE—3001(E)/Code as per SWAYAM/NPTEL-MOOC-3	**3L-1T-0P**	**4**

Guideline

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Courses delivered through SWAYAM/NPTEL-MOOC courses are available free of cost to the learners, however learns wanting SWAYAM/NPTEL-MOOC courses Certificate should register for the final proctoral exams that come at a fee and attend in-person at designated center on specified dates.

In perspective of the need to create synergist between the salient features of 'anytime anywhere format of e-learning and traditional classrooms chalk and talk method to develop a unique content delivery mechanism which is responsive to learners' needs and ensuring seamless transfer of knowledge across geographical boundaries. RKDF University proposes to incorporate MOOC courses delivered by SWAYAM/NEPTEL portal. Student can opt any courses from SWAYAM/NPTEL-MOOC courses as an elective Base on following condition

1. He/She has consulted and inform to the departmental coordinator regarding courses selection and mark obtain to him. No Subjected should be repeat from courses which is offer and courses which are selected.

2. He/She can opt from only courses which are not available from University.

3. Only enrollement in the courses and learning from it is free as this is a project supported by MHRD GOI.

4. To get Certificate from SWAYAM-NPTEL. The learner has to separately register and appear for a proctored exam at a fee as decided by host institution/Mooc of Rs. 1000/- Which be bearded by Student only.

5. If student successful appear in exam and obtain certificated of courses of 12 week course to earn 3 credits(As recommended by NPTEL) to suffice for required credit for the award of his/her degree.

6. The NPTEL recommended that an additional 1 credit may be awarded given by University. This credited can be given based on attendances, assignment and tuitorial work submitted by the students.

7. All students would be motived to take courses online with SWAYAM/NPTEL every semester to better prepared for the future so that they can cultivate the habilt of self-study online. The University propose to implement MOOC courses under Department. The University also made provision if student unable to get certificate He/She still be able to appear in the department elective courses and complete there degree

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Design of Prestressed Concrete Structures**	**MTSE— 3002(A)**	**3L-1T-0P**	**4**

Course Objective

After studying this subject, the students will be able to:

CO1: Understand the concept of prestressed concrete and various types of losses in prestress.

CO2: Analyze and design prestressed concrete sections.

CO3: Analyze and design prestressed concrete members for shear and torsion.

CO4: Analyze and design prestressed concrete beams.

CO5: Analyze and design of special prestressed concrete structures.

Course Contents

Unit 1: Introduction, Prestressing Systems and Material Properties

Introduction, Types of Prestressing, Pre-tensioning and Post-tensioning Systems, Concrete and steel in prestressed members, Pressure line or Thrust line, Concept of load balancing, stresses in Tendons, cracking moment, types of losses in prestressed concrete.

Unit 2: Analysis and design of prestressed concrete sections

Analysis and design of Members under Axial load, Analyses at Transfer and at Service, Analysis and design of Members under Flexural loads for various types of sections, Calculation of Deflection, Calculation of Crack Width, Cracking moment, Kern points, Analysis and design of Members under compression and bending, Detailing Requirements as per relevant codal provisions.

Unit 3: Analysis and design for shear and torsion

Analysis and Design for Shear, Types of Cracks, Stress in Uncracked and uncracked Beam, Analysis and Design for Torsion, Stresses in an Uncracked Beam, Pure Torsion. Transfer of Prestress, End zone reinforcement, anchorage zone reinforcement, Detailing requirements as per relevant codal provisions.

Unit 4: Analysis and design of beams

Limit state design philosophy, design loads and strengths, serviceability criteria, Design of pretensioned, post-tensioned and Partially prestressed beams, Effects of prestressing indeterminate structures, Concordant Tendon Profile, Guyon's theorem, Design of continuous prestressed concrete beams.

Unit 5: Special topics

Composite Sections, One-way Slabs, Two-way Slabs, Circular Prestressing: Prestressed Concrete Pipes, Liquid Storage Tanks, Ring Beams, Design of special structures Poles, Piles, Railway sleepers, etc.

Reference Books

1. Y. Guyen, Prestressed concrete (Vol. I and II), John Willey & Sons
2. N. Krishnaraju, Prestressed concrete, Tata McGraw-Hill
3. T. Y. Lin and N. H. Burns, Design of Prestressed concrete structures, John Willey & Sons
4. S. K. Mallik and A. P. Gupta, Prestressed concrete, Oxford & IBH
5. IS 1343: latest version, Prestressed Concrete - Code of Practice, Bureau of Indian Standards, New Delhi.
6. IS: 3370 (Part 3) latest version Code of practice for concrete structures for storage of liquids - Prestressed Concrete Structures, Bureau of Indian Standards, New Delhi.

The ME students are required to undergo three weeks industrial training during summer vacation of 1st year ME in any Industry/Research Organization /Laboratory/ Engineering Organization/Government Training Institutes/Public Sector undertaking / Academic Institutions of repute. They may also attend Industry/ Job Oriented Courses / Online courses. The students are required to submit a detailed training report and certificate. Evaluation will be done in III semester and is based on report, presentation and subsequent viva voce.

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R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

Second Year (Semester – 3)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Foundation Engineering**	**MTSE—3002(B)**	**3L-1T-0P**	**4**

Courses Outcomes

CO1:-To determine the selection of foundation and sub-soil exploration on various types of soil CO2:-To determine various methods of analysis the bearing capacity of shallow foundation.

CO3:-To determine the selection and mechanics of pile foundation.

CO4:-To determine the important of foundations on problematic soil & Introduction to Geosynthetics.

CO5:-To determine the lateral earth pressure on the retaining wall.

Courses Content

UNIT 1.

Selection of foundation and Sub-soil exploration/investigation: Types of foundation, Factors affecting the selection of type of foundations, Steps in choosing types of foundation based on soil condition. Objectives and planning of exploration program, methods of exploration-wash boring and rotary drilling-depth of boring, Soil samples and soil samplers-representative and undisturbed sampling, Field penetration tests: SPT, SCPT, DCPT. Introduction to geophysical methods, Bore log, report writing.

UNIT 2.

Shallow Foundation: Introduction, significant depth, design criteria, modes of shear failures. Detail study of bearing capacity theories (Prandtl, Rankine, Terzaghi, Skempton, Meyerhof), Bearing capacity determination using IS Code. Settlement, components of settlement & its estimation, permissible settlement, Proportioning of footing for equal settlement, allowable bearing pressure. Bearing capacity from in-situ tests (SPT, SCPT, PLATE LOAD), Factors affecting bearing capacity, Contact pressure under rigid and flexible footings. Floating foundation.

UNIT 3.

Pile foundations: Introduction, Load transfer mechanism, Types of piles and their function, Factors influencing selection of pile, their method of installation and their load carrying characteristics for cohesive and granular soils, Piles subjected to vertical loads- pile load carrying capacity from static formula, dynamic formulae (ENR and Hiley), Pile load test, Pile group: carrying capacity, efficiency and settlement. Negative skin friction.

UNIT 4.

Foundations on problematic soil & Introduction to Geosynthetics: Significant characteristics of expansive and collapsible soils, footing on such soils, Problems and preventive measures. Underreamed pile foundation-its concept, design & field installation. Introduction to geosynthetics-materials, types, functions and uses.

UNIT 5.

LATERAL EARTH PRESSURE: Active, Passive and Earth pressure at rest. Rankine's theory of earth pressure, Earth pressures in layered soils, Coulomb's earth pressure theory, Culmann's graphical method.

RETAININGWALLS: Types of retaining walls- stability of retaining walls against overturning, sliding, bearing capacity and drainage from backfill. Reinforced earth retaining walls.

Reference Book:

1. Murthy, V.N.S., “Text book of Soil Mechanics and Foundation Engineering”, CBS Publishers DistributionLtd.,NewDelhi.2014.
2. Arora, K.R., “Soil Mechanics and Foundation Engineering”, Standard Publishers and Distributors, New Delhi,7thEdition,2017(Reprint).
3. Punmia, B.C., “Soil Mechanics and Foundations”, Laxmi Publications Pvt. Ltd. New Delhi, 16th Edition 2017.
4. Joseph E bowles, “Foundation Analysis and design”, McGraw Hill Education, 5th Edition, 28th August 2015.
5. Shamsher Prakash et al, “Analysis, Design of foundations and Retaining Structures” Sarita Prakashan.
6. Murthy, V.N.S., "Advanced Foundation Engineering", CBS Publishers and Distributors
7. Coduto D.P., Foundation design; principles and practices, Pearson Publication

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Bridge Engineering**	**MTSE—3002(C)**	**3L-1T-0P**	**4**

Courses outcome

CO1:- Student will be developing the requirement of various types of bridge super structures such as R.C and steel bridge.

CO2:- Student will be able to analysis & design of R.C. Bridge. It is include various examples

CO3:- Student will be able to analysis & design of Steel Bridge. It is including various examples.

CO4:-Student will be able to analysis & designs of Pier, Abutment and Wing Walls of Bridge,

CO5:-Student will be able to analysis & designs of Foundation and Bearings of Bridge,.

Courses content

UNIT 1: Types of Bridge Super Structures

Introduction and types, temporary bridge superstructures, military bridges, other temporary bridges, permanent bridges, R.C.C. bridges, Pre-stressed concrete bridges, steel bridges, movable steel bridge. Consideration of loads and stresses in road bridges: Introduction, loads, forces and stress, dead loads, bridge loading as per relevant IRC and IRS specifications traffic lanes, foot way, kerbs, railing and parapet loading, impact, wind load, longitudinal forces, Temperature effect of live load on back fill and on the abutment.

UNIT 2: Design of R.C. Bridge

Slab culvert, pipe culvert, T-beam, box culvert bridge super structure, Courbon's theory for load distribution, balanced cantilever bridges, design examples.

UNIT 3: Design of Steel Bridges

Types of steel superstructure, plate girder bridge, truss bridge, wind forces of lattice girder bridge, bracings, arch and bowstring girder bridge, design example.

UNIT 4: Pier, Abutment and Wing Walls

Types of piers and abutments, stability analysis of piers and abutments, design of piers, Forces on piers, stability, abutment, bridge code provision for abutments, wing walls, design examples.

UNIT 5: Foundations and Bearings

Types of bridge foundations and general design criteria, shallow foundations, deep foundations, piles, wells and pneumatic caissons, river training works.

Bearings: functions and types of bearings, necessity of bearings, design of elastomeric bearings, expansion joints, necessity and types of expansion joints, design considerations.

References Books:

1. Victor, D.J., Essential of Bridge Engineering , Oxford & IBH Publishing Co., New Delhi.
2. Rowe, R.E., Concrete Bridge Design , C.R. Books Ltd., London
3. Krishna Raju N, Design of Bridges, Oxford & IBH Publishing Co., New Delhi.
4. Bakht. B and Jaeger, L.C., Bridge Analysis Simplified, McGraw Hill Book Co. Inc.
5. Ponnuswamy, S., Bridge Engineering, Tata McGraw Hill, New Delhi.
6. Bakht, B. and Jaeger, L.G., Bridge Deck Analysis Simplified, McGraw Hill International Edition, Singapore
7. Aswani M.G., Vazirani V.N. and Ratwani M.M., Design of Concrete Bridges,

Khannapublishers, New Delhi.
8. Hambly E.C., Bridge Deck Behaviour.
9. Sastry V.V., Design of Bridges, Dhanpat Rai & Co
10. Raina V.K., Concrete Bridge Design and Practice, Tata McGraw Hill, New Delhi.
11. Jagadeesh .T.R. and Jayaram.M.A., "Design of Bridge Structures", Prentice Hall of India Pvt. Ltd, Learning Pvt. Ltd., 2013
12. Indian Standard Codes and IRC codes related to bridges

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

Second Year (Semester – 3)

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**RETROFITTING AND REHABILITATION OF STRUCTURES**	**MTSE—3002(D)**	**3L-1T-0P**	**4**

CO1:-apply the various methods of repair, retrofitting and rehabilitation techniques for masonry and concrete structures.
CO2:-evaluate the existing condition of infrastructure, the materials for repair and retrofitting, the maintenance and strengthening techniques .
CO3:-evaluate suitable technique and materials for Seismic retrofitting and design of retrofitted structural components.
CO4:-apply efficient retrofitting and rehabilitation in order to extend the durability of existing structure in a sustainable manner

Unit-1

Introduction and Definition for Repair, Retrofitting, Strengthening and rehabilitation. Physical and Chemical Causes of deterioration of concrete structures, Evaluation of structural damages to the concrete structural elements due to earthquake.

Unit-2:-Damage Assessment:

Purpose of assessment, Rapid assessment, Investigation of damage, Evaluation of surface and structural cracks, Damage assessment procedure, destructive, non-destructive and semi destructive testing systems

Unit-3:-Influence on Serviceability and Durability

Effects due to climate, temperature, chemicals, wear and erosion, Design and construction errors, corrosion mechanism, Effects of cover thickness and cracking, methods of corrosion protection, corrosion inhibitors, corrosion resistant steels, coatings, and cathodic protection.

Unit -4:-Maintenance and Retrofitting Techniques

Definitions: Maintenance, Facts of Maintenance and importance of Maintenance Need for retrofitting, retrofitting of structural members i.e., column and beams by Jacketing technique, Externally bonding(ERB) technique, near surface mounted (NSM) technique, External post-tensioning, Section enlargement and guidelines for seismic rehabilitation of existing building

Unit-5:- Materials for Repair and Retrofitting

Artificial fibre reinforced polymer like CFRP, GFRP, AFRP and natural fiber like Sisal and Jute. Adhesive like, Epoxy Resin, Special concretes and mortars, concrete chemicals, special elements for accelerated strength gain, Techniques for Repair: Rust eliminators and polymers coating for rebar during repair foamed concrete, mortar and dry pack, vacuum concrete, Gunite and Shot Crete Epoxy injection, Mortar repair for cracks, shoring and underpinning.

Course outcomes:

After studying this course, students will be able to:

1. Understand the cause of deterioration of concrete structures.

2. Able to assess the damage for different type of structures

3. Summarize the principles of repair and rehabilitation of structures

4. Recognize ideal material for different repair and retrofitting technique

REFERENCE BOOK:

1) Testing of Concrete in Structure by Bungey (Surrey University Press)
2) Non Destructive Testing by Malhotra & Carino (CRC Press)
3) Corrosion of Steel in Concrete by Broomfield John P. (Taylor & Francis)
4) Concrete Technology, by Neville, A. M. and Brooks, J. J., Prentice Hall
5)Concrete Durability, by Thomas Dyer, CRC Press, Taylor & Francis Group
6) Handbook on Non Destrucive Testing of Concrete; Edited by Malhotra, V. M. and Carino, N. J., CRC Press
7)Composites for Construction, by L. C. Bank, John Wiley & Sons, Inc.
8) ACI 440.2R-08. Guide for the Design and Construction of Externally Bonded FRP Systems for Strengthening Concrete Structures, American Concrete Institute

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH (Structure Engineering)

Second Year (Semester – 3)

Course Content & Grade

Course Code:

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	SWAYAM/NPTEL-MOOC-IV	MTSE—3002(E)/Code as per SWAYAM/NPTEL-MOOC-4	**3L-1T-0P**	**4**

Guideline

SWAYAM/NPTEL-MOOC-IV courses is a programme initiated by Government of India and designed to achieve the three cardinal principles of education via, access, equity and quality. The objective of this effort is to take the best teaching resources to all, including the most disadvantaged SWAYAM seeks to bridge divide for students who have hitherto remained untouched by the digital revolution and have not been able to join the mainstream of the knowledge economy.

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5. If student successful appear in exam and obtain certificated of courses of 12 week course to earn 3 credits(As recommended by NPTEL) to suffice for required credit for the award of his/her degree.

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R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Seminar**	**MTSE-3003**	**0L-0T-4P**	**4**

Students have to give a seminar both in written and oral formats based on the following:

- Recent advancement in the field of Civil Engineering practice and research.
- Scope of further technological advancement in various fields of Civil Engineering.
- Various new techniques, field problems and innovative ideas
- Any other relevant topics

R.K.D.F. UNIVERSITY, BHOPAL

M.TECH **(Structure Engineering)**

Second Year (**Semester – 3)**

Course Content & Grade

Branch	Subject Title	Subject Code	Contact Hours per Week	Total Credits
M.TECH (Structure Engineering)	**Dissertation-1**	**MTSE-3004**	**0L-0T-8P**	**8**

- Identify structural engineering problems reviewing available literature.
- Identify appropriate techniques to analyze complex structural systems.
- Apply engineering and management principles through efficient handling of project